Software Architecture & Protocol Specifications

Technical System Architecture, Pattern & Interface Protocols

1. Architectural Pattern: Model-View-ViewModel (MVVM)

The application architecture strictly follows the MVVM (Model-View-ViewModel) design pattern to ensure complete separation of UI presentation from business logic, high automated testability, and reactive state management.

Architectural Layer	Responsibility & Logic Scope	Primary Components
View Layer (Figma Frames 1–4)	Responsible exclusively for rendering UI components, capturing user touch inputs, and observing ViewModel state streams. Contains zero business logic.	Frame-1 (Home), Frame-2 (Filter Panel), Frame-3 (Results), Frame-4 (Map View)
ViewModel Layer	Holds UI state, processes user interactions, manages 18 backend search parameters, and communicates asynchronously with Repository layer.	FilterViewModel, VenueSearchViewModel, MapNavigationViewModel
Model & Data Layer	Single source of truth for business data. Handles REST API networking, local cache persistence, DTO serialization, and GPS location services.	VenueRepository, LocationManager, APIClient, SearchParametersDTO

2. Backend Search Parameter Protocol (18 Parameters)

The application backend engine relies on 18 standardized query parameters organized into 6 functional domain groups to process multi-criteria venue filtering:

Parameter Group	Key Parameters	Functional Purpose
Location & Map (2)	`radius, bbox`	Defines circular search radius (meters) and bounding box map coordinates.
Text & Content Search (2)	`q / query, search_target`	Text query string and search scope

		(venue name, dish, category).
Category, Food & Cuisine (3)	`shop_type, cuisine_types, dietary_options`	Filters shop category, cuisine origins, and dietary flags (Halal, Vegan, Vegetarian, Gluten-free).
Ratings & Popularity (4)	`min_rating, min_review_count, is_top_search, is_top_comment`	Filters by star ratings (1.0-5.0), total review volume, and popularity trends.
Business Status & Features (3)	`is_open_now, price_level, amenities`	Real-time opening status, price tiers (Student, Mid-range, High), and amenities (Wi-Fi, Power outlets).
Sorting & Pagination (4)	`sort_by, sort_order, page, limit`	Sorting metric (Distance, Rating, Review Count, Smart Score), direction, and page pagination.

3. Communication Protocols & APIs

Protocol Type	Technical Specification	Usage & Purpose
REST API / Data Exchange	HTTPS / TLS 1.3 with Bearer OAuth2 Token	Exchanges search filter payloads and receives structured JSON venue objects.
Location Service	Native Fused Location Provider API	Retrieves real-time device latitude and longitude for distance matrix calculation.
Map Tile Protocol	HTTPS Vector Tile API	Renders interactive background map tiles on Frame-4 with hardware acceleration.
Routing & Polyline API	Directions API (Walking Profile)	Calculates optimal path polyline coordinates between user position and Makarina venue.